MTP F26 · TEAM C · RADIO TEAM

Radio brief: band, module and link budget

Team Santiago, Andrian, Marcos **Version** 0.5 · Tue 13 Oct 2026 **Next session** Fri 16 Oct (week 4)

New in v0.5 (decided 13 Oct, Santiago with Marcos): one omni antenna per box on the Ebyte E01-ML01SP4. Antennas cannot change between the competition modes, and the patches' risks in NM (aiming, the rim corner, tuning) outweighed their +1 dB on MRM. The SP4 has the antenna socket on the module, so the flex omni plugs straight in. Shopping list updated below.

New in v0.4: all four teams agreed to run **every mode at 2.4 GHz**, so 868 MHz is dropped. The radio is now the **Ebyte E01-ML01DP5** (Nordic nRF24L01+ with PA and LNA) at 250 kbps, with **two patch antennas** per box and our own channel in MRM. **Do not order the Adafruit RFM69 or the Molex antennas**; the new shopping list is below. **Order by Fri 16 Oct.**

Every figure carries a tag: `RULE` from the competition rules PDF, `VERIFIED` against a primary source, `ASSUMPTION` to be replaced by a measurement, and `ESTIMATE`.

Bottom line

1. **2.4 GHz is the right band for this competition.** In MRM all teams transmit at once `RULE`. At 868 MHz four teams would share one 250 kHz channel (about a quarter of 67 kbps each). At 2.4 GHz each team gets its own channel, there is no duty-cycle limit, and NM has one common radio for everybody.
2. **The price is margin.** The limit is 10 mW e.i.r.p. (CNAF UN-85 b) `VERIFIED`. With the SP4 and one omni, at 0.75 m: **SRI 70 m +26 dB, MRM 260 m +3.4 dB, NM 100 m hop +9.9 dB**, against +36.7 / +13.9 / +20.5 dB at 868 MHz.
3. **MRM at 260 m is the thin link.** Two levers: antennas 25 cm higher give +5 dB, and a "yes" to Atenea Q3 (100 mW for a frequency-hopping radio) gives +10 dB.
4. **Speed is no longer a problem.** 250 kbps gives ~159 kbps of goodput. The simulator delivers the whole file in 28 s at 5 % loss, and all 10,000 lines even without compression (73 s).
5. **One omni, decided.** Plugged straight into the SP4 (no adapter) and with the SP4's −96 dBm sensitivity, it is only 1 dB behind the two-patch build on MRM, with none of the patches' NM risks. Remaining risk: if T0 measures only ~+2 dBm out of the module, every link loses ~5 dB.
6. **MRM channel plan for the four teams:** RF_CH 24 / 49 / 74 / 82. A neighbouring box 2 m away can arrive 55 dB above our 260 m signal, so the channels sit 25 MHz apart where possible and the two closest teams need 10 m between boxes.

Working baseline sign-off pending

The **E01-ML01SP4** is a genuine Nordic nRF24L01+ with a PA and an LNA, as a 14.85 × 18 mm SMD module with an **IPEX (U.FL) antenna socket**. Datasheet: +20 dBm max, **−96 dBm at 250 kbps**, 120 mA TX, 26 mA RX, 2.0–3.6 V `VERIFIED`. Its lowest power setting gives about **+2 to +7 dBm**,

depending on the PA gain; test T0 measures it before any field test ASSUMPTION. With the 3 dBi peak of the omni that is at most 9.5 dBm e.i.r.p. The driver refuses any higher power setting.

Four radio profiles are generated by `radio/phy_config.py`. Each is checked against the band edges (Carson bandwidth plus ±60 ppm crystal drift per box) and the 10 dBm e.i.r.p. limit for its antenna path.

PROFILE	USE	CHANNEL	RATE	AT SMA	ANTENNA PATH	E.I.R.P.	IN BAND?
TEAM_250K	Our own link: SRI, QM, MRM	74 · 2474 MHz	250 kbps	≤ +7 dBm	omni 3 dBi peak, −0.5 dB	≤ 9.5 dBm	yes · 9.1 MHz to edge
TEAM_1M	SRI option (teams take turns)	74 · 2474 MHz	1 Mbps	≤ +7 dBm	omni 3 dBi peak, −0.5 dB	≤ 9.5 dBm	yes · 8.7 MHz to edge
NM_COMMON	Shared with all teams (below)	80 · 2480 MHz	250 kbps	≤ +7 dBm	omni 3 dBi peak, −0.5 dB	≤ 9.5 dBm	yes · 3.1 MHz to edge

No duty-cycle limit

UN-85 b) has no duty-cycle condition, so the 3-rounds-per-hour limit of 868 MHz is gone. Test as often as we like. The firmware's duty-cycle guard was removed.

The power limit is the only rule to watch. The driver **refuses any setting above RF_PWR −18 dBm**: at full power the module would radiate about 20× the limit.

Firmware ready before the hardware

A portable C driver for the nRF24L01+ passes 124 host checks against a simulated radio, including two radios talking over a simulated channel and the power limit. It needs five board functions (SPI select and transfer, CE pin, microsecond delay and clock).

Frames use dynamic length and the radio's CRC, and go out without the chip's own ACKs, so our ARQ stays in charge. The protocol simulator runs on it: 11 of 11 scenarios pass.

Antenna

One Molex 146153-0100 flex dipole per box (34.9 × 9 mm, 2.4–3.0 dBi peak, > 66 % efficiency), plugged into the SP4's socket, stuck **vertically** as high as possible on a side wall. The same antenna serves SRI, MRM and NM, and Quick Mode too.

The model uses 2 dBi in the box ASSUMPTION; test T5 measures it.

What to buy (order by Fri 16 Oct)

ITEM	QTY	PRICE	WHY
Ebyte E01-ML01SP4 (nRF24L01P + PA/LNA, SMD, IPEX)	6	verify	2 boxes, 2 for the bench, 2 spares. Modules shown at Quick Mode are final, so buy them all from one batch, and check the EU seller and lead time first.
Molex 146153-0100 flex omni, 100 mm cable, U.FL	4	≈ €2–4	The antenna. Check the part is the MHF (U.FL) version, not MHF4.
U.FL-to-SMA-female pigtail	2	€2–3	To measure the module on the spectrum analyser (T0, T13).

Do not buy Ebyte E01C-* parts or no-name "nRF24L01+PA+LNA" boards: Ebyte's own manual says the E01C uses the Si24R1 clone chip. The SP4 is SMD, so it needs a small breakout (or the carrier PCB) from the first bench test. Not needed any more: the DP5, SMA pads and whips.

Antenna: one omni decided 13 Oct

The antennas cannot change between SRI, MRM and NM, so one build has to serve all three. The comparison we decided on, model at 0.75 m (the first row is the decision; the others are the 9 Oct options):

BUILD	E.I.R.P.	SRI 70 M	MRM 260 M	MRM AT 1.0 M	NM 100 M, −10 DB
SP4 + one omni, plugged straight in (decided)	8.5 dBm	+26.0	+3.4	+8.3	+9.9
DP5 + two patches, 4.5 dBi (not chosen)	10.0 dBm	+27.0	+4.4	+9.3	+10.9
DP5 + one omni 2 dBi, via adapter	8.0 dBm	+23.0	+0.4	+5.3	+6.9
One omni straight on the SMA	8.7 dBm	+24.4	+1.8	+6.7	+8.3
One omni, 0 dBi inside the box	6.0 dBm	+19.0	−3.6	+1.3	+2.9
One omni + 100 mW hopping (Atenea Q3 yes)	18.0 dBm	+33.0	+10.4	+15.3	+16.9

Why one omni

- Covers every direction in NM, including the box at the corner of the two rims, with no aiming
- One radio per box, no patch boards, no VNA tuning, off-the-shelf antenna
- Only ~1 dB behind the patches on MRM once it plugs straight into the SP4

What we give up

- A thin MRM margin (+3.4 dB at 0.75 m). The levers: mount the antenna high (+5 dB per 25 cm), and Atenea Q3 (100 mW hopping, +10 dB)
- No sideways rejection of other teams' boxes: the ≥ 10 m spacing of the channel plan still holds, with no extra isolation

Why not the E01-2G4M27D

Its −99 dBm receiver is tempting, but its lowest output is about +9 dBm. Behind a ~3 dBi antenna that is ~12 dBm e.i.r.p., over the 10 mW limit.

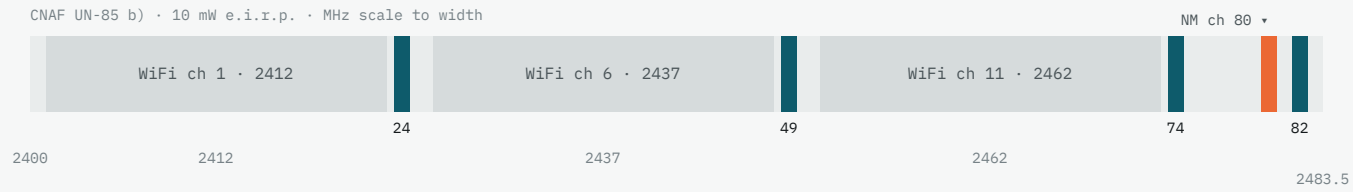
Rules that constrain the radio

- Unlicensed spectrum only, compliant with **CNAF 2017 UN notes**, IEEE C95.1 (RF exposure) and ESD safety. RULE
- **No chip implementing a standard wireless access protocol** (WiFi, Bluetooth, 2G–5G, ZigBee; the Session 1 slide adds NB-IoT and **LoRa**). The nRF24L01+ has none of them. RULE
- **Nothing may stick out of the 15 × 15 × 7 cm IKEA 365+ box** except buttons, switches and connectors, so the antennas go inside. The box sits on a support of **about 70 cm**. RULE
- **Modules shown at Quick Mode are final.** Antennas, batteries and mechanics may still change. RULE
- Distances: SRI **70 m**, teams one after another, 2 minutes; MRM **~260 m, all teams at the same time**, 2 minutes; NM relays along the South and East rims with no line of sight end to end, **5 minutes**. RULE
- Scoring counts only the **first unbroken run of byte-identical lines** (~10,000 lines of ~100 B), so reliability beats raw bit rate. RULE

Spectrum: 2400–2483.5 MHz and our channels

nRF24 channel N sits at 2400 + N MHz, so only **RF_CH 0–83** are legal in Spain; the driver refuses anything higher. Campus WiFi (eduroam) normally uses channels 1, 6 and 11, which leave three

gaps. The MRM plan puts three teams in those gaps, 25 MHz apart, and the fourth at the band edge. Test T14 (campus survey) confirms it before we propose it.



MRM channel, one per team (RF_CH) · NM common channel · typical WiFi occupancy (22 MHz)

BAND	NOTE	MAX POWER	ACCESS CONDITION	STATUS FOR US	COMMENT
2400–2483.5 MHz	UN-85 b)	10 mW e.i.r.p.	none (EN 300 440)	our band	No duty cycle. Every team gets its own channel in MRM. Shared with campus WiFi and Bluetooth.
2400–2483.5 MHz	UN-85 a)	100 mW e.i.r.p.; non-hopping 10 mW/MHz	mitigation required (EN 300 328)	ask Atenea	A narrowband radio gains nothing (10 mW/MHz). Only a frequency-hopping design with listen-before-talk could use 100 mW: +10 dB. Atenea Q3.
869.40–869.65 MHz	UN-39	500 mW e.r.p.	≤10 % DC	dropped 9 Oct	Better range, but one 250 kHz channel for four simultaneous teams in MRM and ~3 rounds per hour.
433 / 5725 MHz	UN-30 / UN-130	10 / 25 mW	≤10 % DC / none	not used	433 MHz antenna too long for the box; few cheap non-WiFi parts at 5.8 GHz.

Sources: CNAF 2017, BOE-A-2017-12318 · CNAF 2021, BOE-A-2021-21346.

MRM: one channel per team

Our ARQ makes both ends transmit (data one way, ACKs back), and the four teams' boxes stand close together at each end. Another team's box a few metres away is far louder than our partner 260 m away. Whether we still hear our partner depends on how far apart the channels are. The nRF24's selectivity at 250 kbps is **−50 dB beyond 6 MHz** and **−60 dB beyond 25 MHz** (datasheet, table 8) VERIFIED.

OTHER TEAM'S BOX FROM OUR RECEIVER	2 M	5 M	10 M	20 M
Interferer above our 260 m signal, omni antennas (our build)	55 dB	47 dB	41 dB	35 dB
Same with patches side by side (~10 dB rejection each; not our build)	35 dB	27 dB	21 dB	15 dB

CHANNEL PAIR	SPACING	SELECTIVITY	WHAT IT NEEDS
24 / 49 · 2424 / 2449 MHz	25 MHz	−60 dB	safe even at 2 m
49 / 74 · 2449 / 2474 MHz	25 MHz	−60 dB	safe even at 2 m
74 / 82 · 2474 / 2482 MHz	8 MHz	−50 dB	≥ 10 m between these two teams' boxes (9 dB to spare)

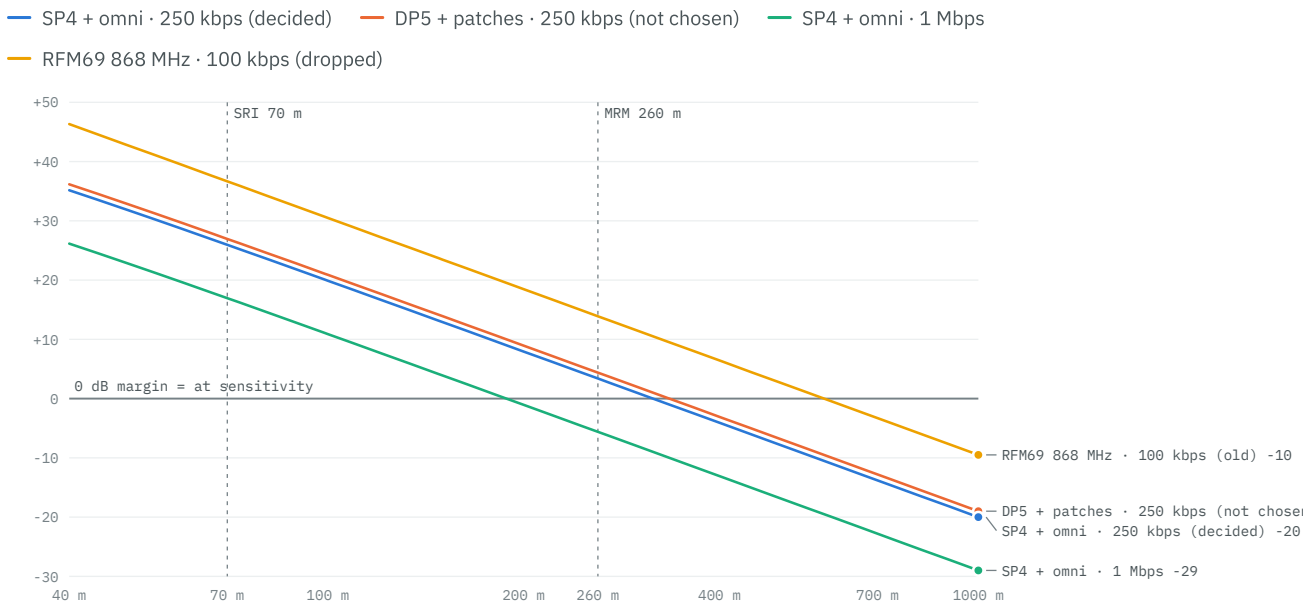
Proposal for the inter-team meeting; we ask the organisers for ≥ 10 m between teams at each end anyway. Test T15 checks it with three of our own boxes.

Link budget

This uses the course method: Maxim AN5142 with the course correction (no ½ factor in the flat-earth formula) and the course spreadsheet's worst-case ground bounce (a = 1, |cos|). Inputs:

- antennas at 0.75 m (70 cm support plus position in the box) ASSUMPTION
- −1 dB container loss and 0 dB multipath on open campus paths ASSUMPTION
- flex omni 2 dBi in the box, cable −0.5 dB, in both directions; +7 dBm out of the module ASSUMPTION
- receiver sensitivity from the E01-ML01SP4 datasheet (−96 dBm at 250 kbps) VERIFIED

FLAT-EARTH LINK MARGIN VS DISTANCE · DB ABOVE RECEIVER SENSITIVITY



▶ Show as table

SCENARIO	EIRP DBM	RX DBM	SENS. DBM	MARGIN DB	WITH −20 DB MULTIPATH
SRI 70 m | E01-ML01SP4 | 250 kbps, omni · decided	8.5	-70.0	-96.0	+25.9	+6.0
SRI 70 m | E01-ML01SP4 | 1 Mbps, omni	8.5	-70.0	-87.0	+16.9	-3.0
MRM 260 m | E01-ML01SP4 | 250 kbps, omni · decided	8.5	-92.6	-96.0	+3.4	-16.6
NM hop 100 m | E01-ML01SP4 | 250 kbps, -10 dB obstr., omni · decided	8.5	-86.1	-96.0	+9.9	-10.1
MRM 260 m | E01-ML01DP5 | 250 kbps, patches (not chosen)	10.0	-89.6	-94.0	+4.4	-15.6

Full AN5142 rows are in `radio/MTP-F26_TeamC_Radio_LinkBudget.xlsx` (scenarios S21–S23 are the decided 2.4 GHz baseline; S16–S20 the patch option); every cell is a live formula and inputs are in blue. AN5142's default −20 dB multipath is for cluttered paths. Tests T4 and T5 on campus will tell us which number to use.

Antenna height is the biggest lever

At 0.75 m the ground reflection takes over beyond ~18 m at 2.4 GHz. Past that point loss grows 40 dB per decade and hardly depends on frequency: $P_R \approx EIRP \cdot G_R \cdot (h_1 h_2)^2 / d^4$.

Raising both antennas from 0.75 to 1.0 m adds **+5 dB** on MRM. Mount the omni as high on the wall as the box allows. Whether "about 70 cm" allows a taller support is Atenea Q4.

The nRF24 has no RSSI

It only has one bit (RPD) that says "above −64 dBm at the chip", about −74 dBm at the SP4's antenna socket. Range tests therefore measure packet loss, and levels come from the spectrum analyser or the model. RPD still works as listen-before-talk and for the campus channel survey.

Datasheet limits that matter

- **32 B maximum payload:** our frames shrank from 64 B to 32 B (28 B of data). `VERIFIED`
- **Variable-length frames need the chip's auto-ACK feature on,** so the driver sends every frame as "no ACK": no chip ACKs or retries on air.
- **PA always in line:** no bypass, so the lowest output is +2 to +7 dBm; the driver caps the setting there.

Throughput: does the file fit in 2 minutes?

About 1 MB in 120 s needs **66.7 kbps** of useful throughput. The model assumes selective-repeat ARQ with one ACK per 16 frames and 85 % efficiency for the micro and USB `ASSUMPTION`. The simulator runs the real protocol code over a modelled channel with 5 % random loss, deflate compression, and the nRF24 airtime from the driver.

CONFIGURATION	AIR RATE	GOODPUT (MODEL)	WHOLE FILE (SIM, 5 % LOSS)	LINES, NO COMPRESSION
SP4, 32 B frames (decided)	250 kbps	158.7 kbps	28 s	10,000 (73 s)
SP4, 32 B frames, SRI option	1 Mbps	594.1 kbps	13 s	10,000
RFM69 868 MHz, 64 B frames (dropped)	100 kbps	67.3 kbps	38 s	9,643
RFM69 868 MHz shared by 4 teams in MRM (dropped)	~25 kbps	~17 kbps	–	~2,500

> At 2.4 GHz each team keeps its full rate in MRM, because it has its own channel. That, not range, is why the band switch pays off.

Module shortlist

Read from the primary datasheets on 9 Oct 2026. Prices and EU lead times are still **TO VERIFY**.

MODULE	CHIP	OUT AT THE CONNECTOR	SENSITIVITY	CONNECTOR	PROTOCOL-BAN RISK	NOTES
Ebyte E01-ML01SP4 decided	nRF24L01P + PA/LNA	~+2 to +20 dBm	−96 @250k (datasheet)	IPEX on the module	low–med	Genuine Nordic chip per Ebyte, FCC/CE. SMD 14.85 × 18 mm, 1.27 mm pitch. The flex omni plugs straight in.
Ebyte E01-ML01DP5 not chosen	nRF24L01P + PA/LNA	~+2 to +20 dBm	~−96 @250k	SMA	low–med	Same radio with pin headers. Would need an SMA-to-U.FL adapter (~0.5 dB) for the flex omni.

MODULE	CHIP	OUT AT THE CONNECTOR	SENSITIVITY	CONNECTOR	PROTOCOL-BAN RISK	NOTES
Ebyte E01-2G4M27D	nRF24L01P + 27 dBm PA	≥ +9 dBm	−99 @250k	SMA	low–med	Its ~+9 dBm floor behind a ~3 dBi antenna is ~12 dBm e.i.r.p., over the limit. Out.
Bare nRF24L01+	nRF24L01+	−18 to 0 dBm	−94 @250k	PCB / none	low–med	Legal and simple, but 7 dB less than the E01 at the same e.i.r.p. limit.
TI CC2500 modules	CC2500	+1 dBm	−87 @250k	PCB	low	Weaker receiver, no ready module with a connector or PA. Out.
E01C-*, no-name PA+LNA boards	Si24R1 clone	unknown		SMA	exclude	Ebyte's own E01C manual names the Si24R1.
SX1280 / SX1281	LoRa 2.4 GHz	—			exclude	LoRa is named on the Session 1 slide.
Radio SoCs (nRF52, ESP32, CC2652…)	—	—			exclude	A second micro and Bluetooth/ZigBee stacks.

Decisions and who they touch

DECISION	OWNER	DEPENDS ON	AFFECTS	STATUS
D-R1b Band: 2.4 GHz for every mode, 250 kbps	All teams + Radio	—	Everything radio	agreed by the teams, 9 Oct
D-R2c Radio module (frozen at Quick Mode)	Radio → handed to electronics	T0 output power, T2 sensitivity	PCB, BoM, power supply	decided 13 Oct: E01-ML01SP4 order by Fri 16 Oct
D-R3 NM radio shared by all teams	Radio + protocols	Other teams' modules	NM bonus (+2 to +2.5)	draft v0.2 ready send to teams; freeze by week 6
D-R4 MRM channel per team	Radio + other teams	Campus survey T14	MRM (+1)	draft: 24 / 49 / 74 / 82
D-R5c Antenna: one flex omni (Molex 146153-0100)	Radio	Box inside height, T5	Mechanics, carrier PCB	decided 13 Oct T5 measures the in-box gain

> **Revisit D-R2c if** T0 measures more than ~+7.5 dBm at the lowest setting (the e.i.r.p. limit with a 3 dBi antenna), or T2 shows a clone or a weak receiver. If T0 measures only ~+2 dBm, the module is legal but every margin drops ~5 dB: then raise the antenna and push Atenea Q3.

Handoff to electronics

One **E01-ML01SP4** per box. SMD, 10 pads at 1.27 mm: 1 VCC, 2 CE, 3 CSN, 4 SCK, 5 MOSI, 6 MISO, 7 IRQ, 8–10 GND (Ebyte manual §3). No traces under the module on the top layer.

PIN	AT THE MICRO	WHAT MATTERS
`VCC, GND`	supply	**2.0–3.6 V only**, never the battery rail (a Li-ion cell reaches 4.2 V). Use a dedicated 3.3 V LDO for the radio (≥ 300 mA), and 100 µF + 10 µF + 100 nF at the module. 120 mA in TX, 26 mA in RX.
`SCK MOSI MISO`	SPI master	Mode 0, MSB first, ≤ 10 MHz.
`CSN`	GPIO out	Chip select, active low.
`CE`	GPIO out	High = receive; a ≥ 10 µs pulse sends one frame.
`IRQ`	GPIO in, optional	Active low: frame sent / frame received. The driver polls today.
`IPEX`	—	The flex omni plugs in here. Keep the module near the antenna wall and away from the battery, USB and the 5 V boost.

Proposed Pico 2 pins: SPI0 GP16/18/19, CSN GP17, CE GP20, IRQ GP21. The SP4 needs a small breakout (or the carrier PCB) from the first bench test, because the Quick Mode module is final. Needed back from electronics: OK on the module and the radio LDO, who makes the breakout, the pin map, a power check, and the box's inside height.

Shared NM radio settings: draft for all teams

Plain nRF24L01+ register settings, so any team's nRF24 board can join with a few register writes and no shared code.

SETTING	VALUE	WHY
Channel	`RF_CH 80 · 2480 MHz`	Clear of WiFi 1/6/11 and of the four MRM channels
Rate	`250 kbps`	Best sensitivity and selectivity; NM moves only ~0.5 kB
Address	`5 B · C2 5A 9D 3E 71`	Different from every team's own address
Frames	dynamic length 1–32 B, hardware CRC-16	Same on every nRF24L01+
ACKs	none on air: send with "no ACK", retries 0	Chip ACKs from several receivers would collide
Power · polarisation	≤ 10 dBm e.i.r.p. · vertical	Legal limit; mismatched polarisation costs 10–20 dB
Channel access	listen before talk with RPD, random back-off	The nRF24 has no carrier sense in hardware

Interop test before week 6: every pair of boxes exchanges 1 000 frames at 10 m on RF_CH 80; pass is ≥ 99 % with a valid end-to-end CRC. Full proposal: `radio/NM_PHY_PROPOSAL.md`.

Test plan

No duty-cycle limit, so tests can be repeated freely. Stay at RF_PWR −18 dBm until T0. For each run, record place, profile, channel, distance, heights, antenna, box open or closed, pad, and frames sent and received.

TEST	WHAT	PASS	WHEN
T0	Bring-up, then **conducted output power** at the lowest setting on the spectrum analyser (U.FL-to-SMA pigtail)	Driver starts; output ≤ +7.5 dBm, so e.i.r.p. ≤ 10 dBm with the 3 dBi antenna. **Gate for any field test.**	modules arrive
T2	Sensitivity at 250 kbps and 1 Mbps through a lab step attenuator	1 % packet loss at ≤ **−92 dBm** (250 kbps). **Gate for the module.**	before Quick Mode
T14	Campus 2.4 GHz survey with the RPD bit, channels 0–83, at each site	Our MRM channels and the NM channel busy < 5 %	before the inter-team meeting
T4	Range, open box, 10 / 70 / 150 / 260 m; then one 2-min transfer at 70 m	≤ 1 % loss to 150 m; all 10,000 lines	after T2
T13	Spectrum analyser: carrier offset, occupied bandwidth on channel 82, e.i.r.p.	Below 2483.5 MHz; ≤ 10 dBm. Plot in the ANNEX.	book the lab
T5	Closed box; antenna high vs middle of the wall; box rotated 0 / 90 / 180°	≤ 1 % loss at 260 m	after T4
T15	MRM near-far with a third box on channel 82 and 49 at 2 m and 10 m	≤ 1 % loss on the 260 m link	after T5

EXPECTED (MODEL)	10 M	70 M	150 M	260 M
SP4 + omni · level	−56 dBm	−70 dBm	−83 dBm	−93 dBm
SP4 + omni · margin	+40 dB	+26 dB	+13 dB	+3 dB

Next: to Fri 16 Oct

ITEM	DATE ASG.	ASSIGNED TO	DESCRIPTION	STATUS	DUE DATE	END DATE	RESULT
R7	09/10/26	RT + PM	Band, module and antenna decisions	Done	16/10/26	13/10/26	2.4 GHz (09/10); SP4 + one omni (13/10)
R12	13/10/26	RT → ET	Order 6 × E01-ML01SP4, 4 × Molex 146153-0100, 2 × U.FL-to-SMA pigtails	Open	16/10/26		
R13	09/10/26	Andrian	Campus 2.4 GHz survey (T14) at the SRI, MRM and NM sites	Open	16/10/26		
R14	09/10/26	RT	MRM channel plan + NM radio settings v0.2 to the other teams' leaders	Open	16/10/26		
R9	13/10/26	RT → ET	Electronics handoff v0.3 reviewed; pin map, radio LDO, SP4 breakout and box height back	Open	16/10/26		
R11	04/10/26	RT	Tests T0 and T2 done; D-R2b confirmed or reversed	Open	before QM		

Questions to ask

Atenea forum (posted by the project leader)

- **Allowed chips.** Is the nRF24L01+ on the Ebyte E01-ML01SP4 allowed? Its packet format is Nordic-proprietary and it has no Bluetooth.
- **2.4 GHz power.** May a radio that hops over ≥ 15 channels with listen-before-talk use the 100 mW of UN-85 a), or must every team stay at 10 mW (UN-85 b)? Worth +10 dB on MRM.
- **Support height and box.** Is the support fixed at 70 cm? May the box stand on its 15 × 7 cm edge?
- **Cooperation.** May teams agree a common NM radio and an MRM channel split, sharing settings but not code?

Consultants

- **Anna Umbert** (spectrum regulation): what would a frequency-hopping design need to show for 100 mW under EN 300 328? What e.i.r.p. evidence does she expect in the ANNEX?
- **Concepción Santos / Carles Puente** (antennas): placing a flex omni inside a plastic box on a 70 cm support; measuring its in-box gain at the lab; effect of the battery, the foil and hands.
- **Paradells / Perelló** (protocols, with the protocols team): one channel per team in MRM and listen-before-talk in NM.

Files: `radio/MTP-F26_TeamC_Radio_LinkBudget.xlsx` (hand-in link budget, live formulas) · `radio/linkbudget.py` (same model, plus the MRM near-far printout) · `radio/phy_config.py` (radio profiles and checks) · `radio/fw/` (nRF24 driver and tests) · `radio/DECISIONS.md`, `HANDOFF_ET.md`, `NM_PHY_PROPOSAL.md`, `TEST_PLAN.md`, `ATENEA_QUESTIONS.md`. Repo: github.com/SantiBarrios2002/mtp-f26-teamc.

Primary sources: competition rules PDF (MTP F26, Network Upgrade III + Sustainability); Session #1 slides; AN5142 with the course annotations; CNAF 2017/2021 (BOE); nRF24L01+ Product Specification v1.0; Ebyte E01-ML01SP4, E01-2G4M27D and E01C manuals; Molex 146153 and Taoglas FXP73 antenna listings.